Supplementary Evidence for AMRL

S1. Complete five-seed condition summaries

Every row in Table S1 uses training seeds 0–4 and a 100,000-step base-training budget. Each seed score is the mean of 20 separately seeded deterministic evaluation episodes. SD denotes the sample standard deviation across the five training seeds, not across evaluation episodes. CartPole uses a declared 1000-step episode cap; bsuite values are raw episode returns for the specified `/0` instances; and `memory_len/0` has memory length 1. Main results and their shared transfer references are counted once, yielding 265 distinct base-training runs in 53 complete condition groups. No incomplete exploratory group is pooled with these comparisons.

Table S1: All complete condition groups ($n = 5$ training seeds per row).

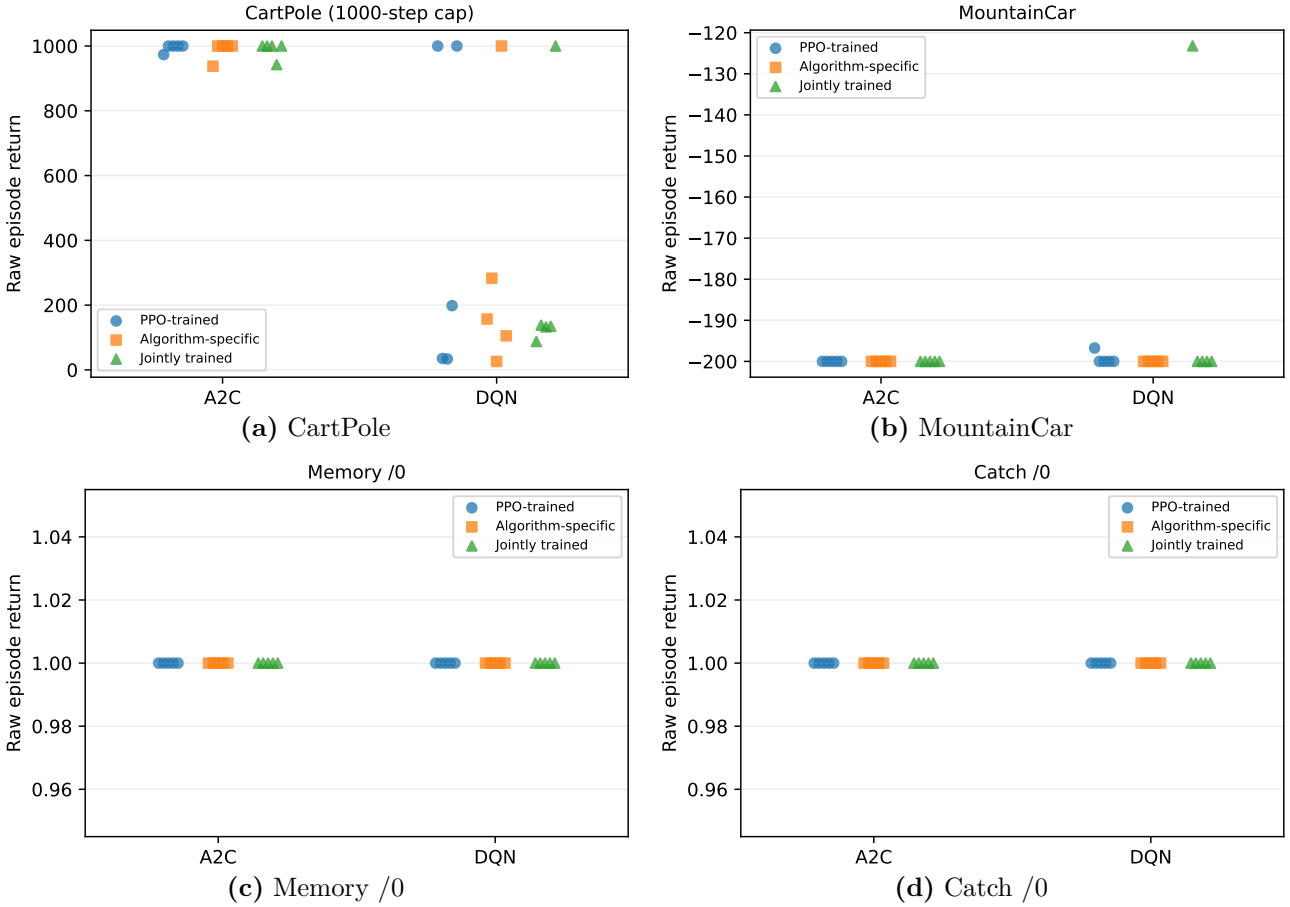
Task	Base	Condition	Mean	SD
CartPole (1000)	A2C	PPO-pooled AMRL	994.68	11.90
CartPole (1000)	A2C	Jointly trained	988.53	25.65
CartPole (1000)	A2C	Algorithm-specific	987.55	27.84
CartPole (1000)	A2C	Static (TPE)	838.82	360.41
CartPole (1000)	DQN	PPO-pooled AMRL	453.50	503.34
CartPole (1000)	DQN	Jointly trained	298.69	392.59
CartPole (1000)	DQN	Algorithm-specific	314.31	394.52
CartPole (1000)	DQN	Static (TPE)	284.43	403.64
CartPole (1000)	PPO	PPO-pooled AMRL	762.71	329.73
CartPole (1000)	PPO	Double bounds	749.19	345.07
CartPole (1000)	PPO	Half bounds	833.94	356.75
CartPole (1000)	PPO	No actuation	918.78	128.12
CartPole (1000)	PPO	Double interval	770.73	289.84
CartPole (1000)	PPO	Half interval	839.48	349.04
CartPole (1000)	PPO	Matched MLP	861.65	298.58
CartPole (1000)	PPO	Reward only	711.11	395.67
CartPole (1000)	PPO	Static (TPE)	918.78	128.12
MountainCar	A2C	PPO-pooled AMRL	-200.00	0.00
MountainCar	A2C	Jointly trained	-200.00	0.00
MountainCar	A2C	Algorithm-specific	-200.00	0.00
MountainCar	A2C	Static (TPE)	-200.00	0.00
MountainCar	DQN	PPO-pooled AMRL	-199.35	1.45
MountainCar	DQN	Jointly trained	-184.64	34.35
MountainCar	DQN	Algorithm-specific	-200.00	0.00
MountainCar	DQN	Static (TPE)	-199.82	0.40
MountainCar	PPO	PPO-pooled AMRL	-200.00	0.00
MountainCar	PPO	Static (TPE)	-200.00	0.00
MountainCar	PPO	Target-trained	-200.00	0.00
MountainCar	PPO	Source: CartPole	-200.00	0.00
Catch /0	A2C	PPO-pooled AMRL	1.00	0.00
Catch /0	A2C	Jointly trained	1.00	0.00
Catch /0	A2C	Algorithm-specific	1.00	0.00
Catch /0	A2C	Static (TPE)	1.00	0.00
Catch /0	DQN	PPO-pooled AMRL	1.00	0.00
Catch /0	DQN	Jointly trained	1.00	0.00
Catch /0	DQN	Algorithm-specific	1.00	0.00
Catch /0	DQN	Static (TPE)	1.00	0.00
Catch /0	PPO	PPO-pooled AMRL	1.00	0.00
Catch /0	PPO	Static (TPE)	0.74	0.25
Catch /0	PPO	Target-trained	0.90	0.22
Catch /0	PPO	Source: MountainCar	0.90	0.22
Memory /0	A2C	PPO-pooled AMRL	1.00	0.00
Memory /0	A2C	Jointly trained	1.00	0.00
Memory /0	A2C	Algorithm-specific	1.00	0.00
Memory /0	A2C	Static (TPE)	1.00	0.00

Task	Base	Condition	Mean	SD
Memory /0	DQN	PPO-pooled AMRL	1.00	0.00
Memory /0	DQN	Jointly trained	1.00	0.00
Memory /0	DQN	Algorithm-specific	1.00	0.00
Memory /0	DQN	Static (TPE)	1.00	0.00
Memory /0	PPO	PPO-pooled AMRL	1.00	0.00
Memory /0	PPO	Static (TPE)	1.00	0.00
Memory /0	PPO	Target-trained	1.00	0.00
Memory /0	PPO	Source: CartPole	1.00	0.00

S2. Cross-algorithm references and seed-complete control trajectories

S2.1. Cross-algorithm controller references

Figure S1 reports the complete cross-algorithm reference set used in the revision. PPO-trained, algorithm-specific, and jointly trained controllers are shown for A2C and DQN on all four tasks. Every point is one training-seed evaluation mean; diamonds denote the corresponding five-seed group means. Favorable, unfavorable, and tied seeds are all retained.



(a) CartPole (b) MountainCar

(c) Memory /0 (d) Catch /0

Figure S1: Complete cross-algorithm reference groups. Points show individual training-seed evaluation means and diamonds show group means. PPO-trained, algorithm-specific, and jointly trained controllers receive the same total of 36 meta-episodes, with different exposure per algorithm. All references are displayed for A2C and DQN on all four tasks. Identical bsuite scores reflect observed ceilings and should not be interpreted as established equivalence.

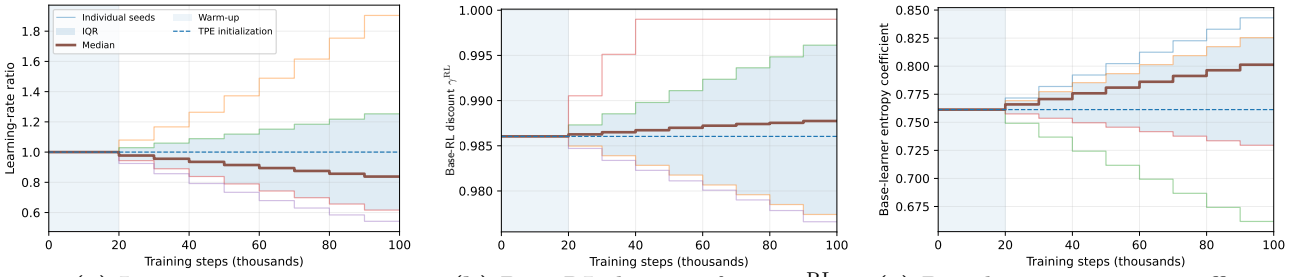
S2.2. Seed-complete projected hyperparameter trajectories

The source-data archive contains the actual CartPole–PPO AMRL control log for every one of the five main-comparison seeds. Each record provides the TPE initialization and eight intervention records at 20,000–90,000

training steps, including the projected hyperparameters actually supplied to the base learner. We therefore use these logs rather than the earlier single-run representative trajectory display.

Figure S2 shows all five trajectories and overlays the pointwise median and interquartile range (IQR). The IQR is used only as a descriptive visual summary; with five seeds it is not a population confidence interval. Learning rate is expressed relative to its TPE initialization, whereas the base-RL discount factor γ^{RL} and the PPO base-learner entropy coefficient remain in their original units. The 0–20,000-step interval is the prescribed warm-up period.

The five final evaluation returns represented in the same trajectory set are seed 0: 1000.00, seed 1: 1000.00, seed 2: 327.45, seed 3: 486.10, seed 4: 1000.00. Thus, the trajectory display intentionally contains both strong and unfavorable final outcomes. Its purpose is to make the realized control path auditable, not to argue that bounded interventions are necessarily beneficial.



(a) Learning-rate ratio **(b)** Base-RL discount factor γ^{RL} **(c)** Base-learner entropy coefficient

Figure S2: Seed-complete projected hyperparameter trajectories for the five CartPole–PPO AMRL runs used in the main paired comparison. All individual seed trajectories are retained; the thicker step curve is the pointwise median and the shaded envelope is the descriptive IQR. The lightly shaded 0–20k interval denotes warm-up, and the dashed horizontal reference denotes the TPE initialization. Step segments hold the projected value until the next eligible intervention. Learning rate is normalized by its initialization; base-RL discount factor γ^{RL} and entropy coefficient use their original units. The trajectories expose the realized sparse, projected control behavior, but they do not establish beneficial adaptation, convergence, or closed-loop stability. The discount-factor panel label was revised with assistance from OpenAI Codex.

S3. Cost accounting and retained exploratory records

The 70 retained controllers account for 252 million inner-training interactions. The 12 shared TPE studies account for 180 million training and 6,899,973 evaluation interactions. The 265 deployments account for 26.5 million training and 1,504,269 evaluation interactions. Using the unrounded stage-level records, the completed stage timers sum to 17.946 hours. Rounded component totals reported in the main manuscript may not sum exactly to this value because each component is rounded independently. These totals are neither continuous elapsed duration nor processor-utilization time. TPE timing includes candidate evaluation, while final deployment evaluation is reported separately.

The original expanded queue also produced 42 completed exploratory deployments and twelve additional controllers not needed by the focused evidence set. Their completed meta-training timers total 1.714 hours, and deployment plus evaluation timers total 0.175 hours. They remain in the accompanying exploratory source table with their actual seed coverage; no missing values are imputed. These records are excluded from the main cost figure, whose scope is the retained 265-run experiment set. Interrupted controller-training attempts have no complete timing records; their partial costs are unknown.

S4. Source data, reproducibility, and interpretation

This supplementary PDF is provided through the journal's submission system. The source-data archive `source_data.zip` and preserved implementation snapshot `reference_code.zip` are available through `https://github.com/yhily/AMRL-Meta-Net3`. The source-data README lists each file, and `SHA256SUMS.txt` provides checksums. Formal and additional exploratory records are stored separately; incomplete records are not used to fill missing seeds. Trained model weights are not included.

The source-data package provides the 265 seed-level records, 5,300 individual evaluation episodes, all paired-test results, the 70 unique controller cost records, twelve TPE cost records, and the deduplicated cost ledger. Figure S2 is generated directly from `plot_records.json`; the extracted values used for the figure are additionally

provided as `cartpole_ppo_amrl_seed_trajectories.csv`, so the displayed control paths can be audited without retraining.

The twelve main comparisons use Holm adjustment. The remaining paired contrasts are exploratory, and their unadjusted intervals do not provide simultaneous coverage. All-zero differences yield an observed tie; zero-width empirical intervals and undefined standardized effects must not be interpreted as equivalence tests. The source data retain unfavorable and tied outcomes, and figure layout does not select seeds by performance.

The five runs in Figure S2 are exactly the five CartPole–PPO AMRL training seeds used in the main paired comparison. Figure S2 is therefore a mechanistic companion to the paired-return analysis rather than a separate sample. Inferential performance claims remain based on the paired evaluation outcomes reported in the manuscript and source data.